Discussion week 3

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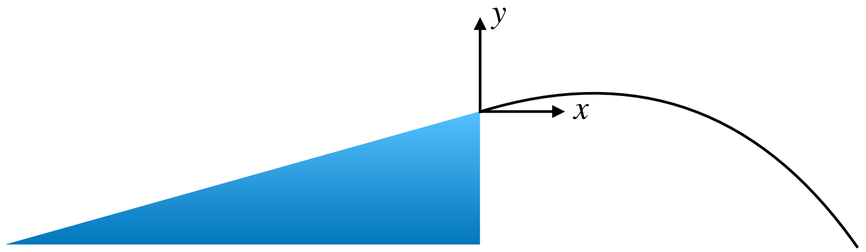
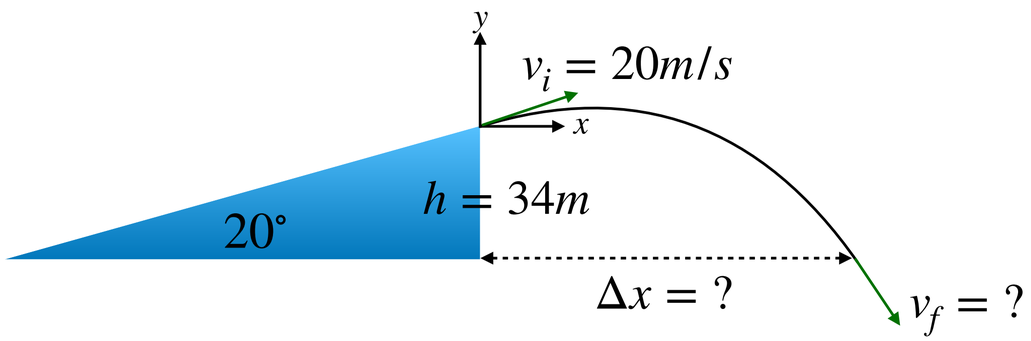
1. Projectile Motion

1. A stunt man drives a car at a speed of 20 m/s off a 34-m high cliff. The road leading to the cliff is inclined upward at an angle of 20 deg.



a. How far from the base of the cliff does the car land?
b. What is the car's impact speed?

1. Projectile Motion



a. How far from the base of the cliff does the car land?

Let $y(t) = v_i \sin 20° t - \frac{1}{2} g t^2 = -34m$, then the time of flight is around 3.42 s

Therefore, $\Delta x = v_i \cos 20° t = 64.3m$

c. What is the car's impact speed?

$$v_f = \sqrt{v_{fx}^2 + v_{fy}^2}$$

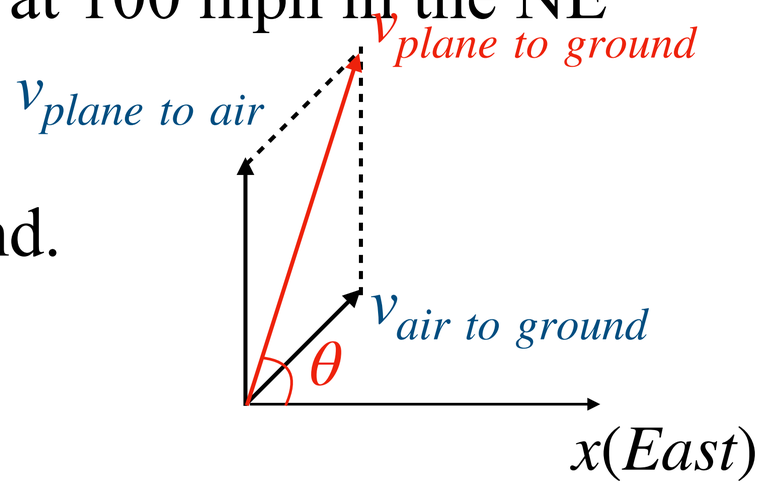
$$v_{fx} = v_{ix} = v_i \cos 20°$$

$$v_{fy} = v_{iy} - gt = v_i \sin 20° - gt$$

$$v_f = 32.7m/s$$

2. Relative Motion

2. An airplane is pointing North and flying at 450 mph relative to the air around it. The jet stream in which the plane is traveling is moving at 100 mph in the NE direction.



a. Draw the airplane's velocity vector relative to the ground.

b. What is the magnitude and direction of the airplane?

$$v_{tot} = \sqrt{v_x^2 + v_y^2} = \sqrt{(100\cos 45^\circ)^2 + (100\sin 45^\circ + 450)^2} = 525.5(mph)$$

$$\theta = \tan^{-1}\frac{v_y}{v_x} = \tan^{-1}\frac{450 + 100\sin 45^\circ}{100\cos 45^\circ}$$

c. If the airplane increases its speed to 500 mph, how does that affect its velocity in the East-West direction?

3. Circular Motion I

3. A Formula 1 car drives at 40.0 m/s around a 90 degree turn with a radius of 32.5 m.

a. What is the centripetal acceleration felt by the driver?

b. How many g's is this? That is, how many times the acceleration of gravity does the driver feel?

c. In what direction is the acceleration pointing?

d. The next time around the track, the driver accelerates through the turn, starting it at 40.0 m/s and ending it at 42.0 m/s. What is the *total* acceleration felt by the driver? Draw each component and give the total magnitude and direction (e.g. at the halfway point around the turn).

3. Circular Motion I

3. A Formula 1 car drives at 40.0 m/s around a 90 degree turn with a radius of 32.5 m.

a. What is the centripetal acceleration felt by the driver?

$$a = \frac{v^2}{R} = 49.2 m/s^2 = 5g$$

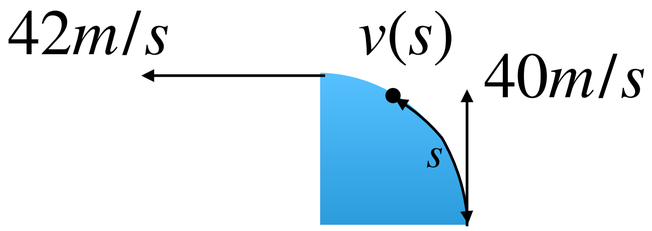
b. How many g's is this? That is, how many times the acceleration of gravity does the driver feel?

3. Circular Motion I

3. A Formula 1 car drives at 40.0 m/s around a 90 degree turn with a radius of 32.5 m.

c. In what direction is the acceleration pointing?

d. The next time around the track, the driver accelerates through the turn, starting it at 40.0 m/s and ending it at 42.0 m/s. What is the *total* acceleration felt by the driver? Draw each component and give the total magnitude and direction (e.g. at the halfway point around the turn).



s : distance from the initial position along the arc

$$a_t = \frac{v_f^2 - v_i^2}{2 \times \frac{1}{2}\pi R} \qquad a_n = \frac{v^2(s)}{R} \qquad v^2(s) - v_i^2 = 2a_t s$$

Therefore, $a_n(s) = \dfrac{v_i^2 + 2a_t s}{R}$, and $a_{tot}(s) = \sqrt{a_t^2 + a_n^2(s)}$

4. Circular Motion 2

4. Astronauts use a centrifuge to simulate the acceleration of a rocket launch. The centrifuge takes 40.0 s to speed up from rest to its top speed of 1 rotation every 1.50 s. The astronaut is strapped into a seat 6.50 m from the axis.

a. What is the astronaut's tangential acceleration during the first 40.0s ?

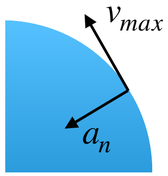
$$a_t = \frac{v_f - v_i}{\Delta t} = \frac{\omega_f R}{\Delta t} = \frac{\frac{2\pi}{T}R}{\Delta t} = \frac{\frac{2\pi}{1.5}6.5}{40} = 0.68 m/s^2$$

b. What is the magnitude of the astronaut's net acceleration 3.0 s after the centrifuge starts speeding up?

$$a_n = v^2/R = (a_t t)^2/R = (0.68 \times 3)^2/6.5 = 0.64 m/s^2$$

$$a_{tot} = \sqrt{a_n^2 + a_t^2} = 0.94 m/s^2$$

c. What is the direction of the net acceleration when the astronaut is rotating at top speed?



$a_t = 0$, and a_n is along radial direction.